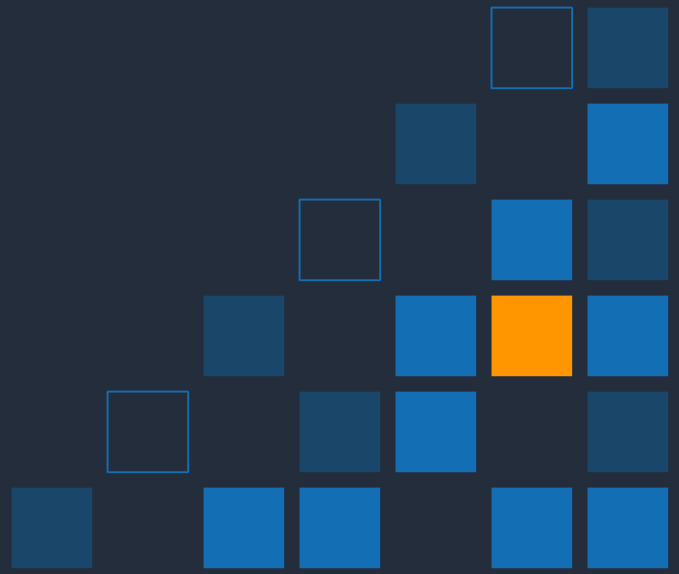
AWS STUDENT BUILDER GROUP · REVA UNIVERSITY

IMPULSE '26

24-Hour National-Level Hackathon

23–24 October 2026 | Rangasthala, REVA University, Bengaluru

MENTORING ROUND · SOLUTION PRESENTATION



Team Introduction

Team Name

[Enter team name]

College / University

[Enter institution name]

Team Members

Product: **DoseWISE** · Better medication management

#	Name	SRN	Branch	Sem	Year	Phone Number
01	*[Full name] (Team Lead)*	*[SRN]*	*[Branch]*	*[Sem]*	*[Year]*	*[Phone number]*
02	*[Full name]*	*[SRN]*	*[Branch]*	*[Sem]*	*[Year]*	*[Phone number]*
03	*[Full name]*	*[SRN]*	*[Branch]*	*[Sem]*	*[Year]*	*[Phone number]*
04	*[Full name]*	*[SRN]*	*[Branch]*	*[Sem]*	*[Year]*	*[Phone number]*

Solution Overview

Solution Summary

DoseWISE is a web platform that connects patients, doctors and family caregivers around **one shared, versioned medication record**.

Doctors prescribe and edit with full history and automatic safety checks. Patients get reminders and log every dose. Caregivers see only what the patient allows. In an emergency, any doctor can open a read-only chart using the patient's **Access ID**.

Built with React + FastAPI and running on AWS in Mumbai (ap-south-1).

Core Value Proposition

One trustworthy medication record that follows the patient.

- Every dose logged: taken, late or missed
- Every prescription change versioned and safety-checked
- Every access consented, time-limited and audited

Key Features

01

Versioned e-prescriptions, safety-checked for duplicates, allergies and **Bedrock AI** screening

02

Reminders and an honest dose log (**taken / late / missed**) the doctor sees live

03

Patient Access ID: 90-day doctor access or a one-time guest emergency view

Target Users

- Adults on long-term medicines from several doctors (diabetes, hypertension, elderly care)
- Family caregivers who help manage medicines
- Doctors, clinics and emergency physicians in India

Problem Validation & Approach

EXISTING GAP

Problem Context

People with chronic conditions take several medicines prescribed by different doctors. Prescriptions live on paper slips and in separate hospital systems, so **no one sees the full, current list**.

The WHO estimates that only about **50%** of patients with chronic disease take their medicines as prescribed (WHO, 2003). Our pilot would measure our own baseline.

Current Limitations

- No shared, versioned medication record across doctors
- Missed doses go unnoticed until the next visit
- Emergency doctors can't see current medicines or allergies
- Reminder apps aren't linked to the prescriber or the caregiver

PROPOSED APPROACH

Proposed Approach

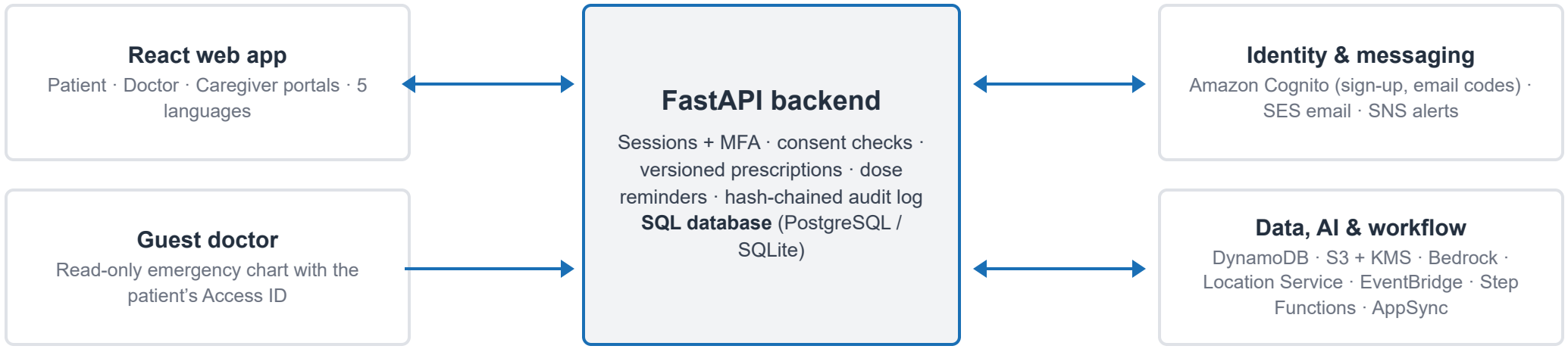
DoseWISE makes the prescription the **single source of truth** and connects everyone to it:

- **Doctors** issue versioned prescriptions; every change is checked for duplicate therapy, documented allergies and AI-screened drug interactions (Amazon Bedrock, shown as unverified)
- **Patients** get reminders and log each dose; late (>60 min) and missed doses reach the doctor and caregiver
- **Consent first**: Access ID for doctors, scoped caregiver access and a read-only emergency view, all audited
- **Built on AWS** in Mumbai: Cognito, DynamoDB, Bedrock, SNS, Location Service, S3 + KMS

Technical Architecture

Architecture Diagram

Every request passes a server-side consent check · region ap-south-1 (Mumbai)

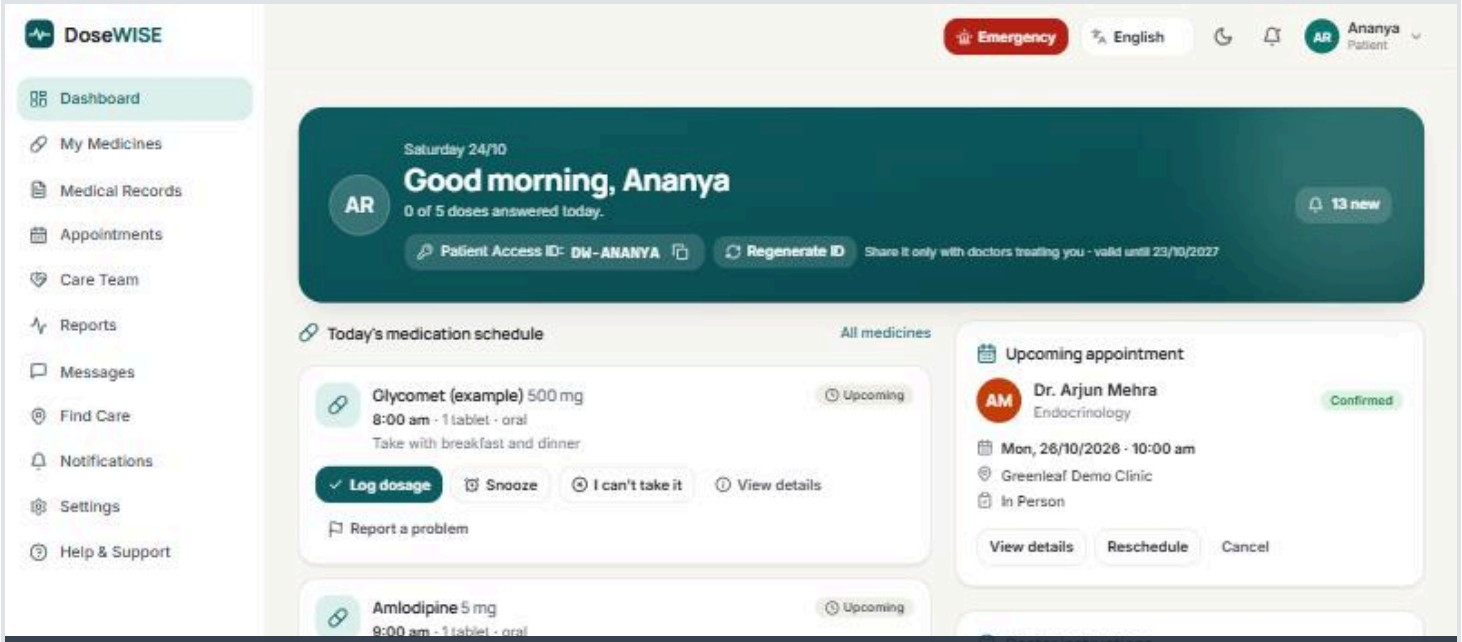


Technology Stack

Web app today; deployable to AWS with the CDK stack in **infra/**

FRONTEND	BACKEND	DATABASE	AI / ML	CLOUD / INFRA	OTHER TOOLS
React 18 · TypeScript · Vite · Tailwind · TanStack Query	Python · FastAPI · SQLAlchemy · Alembic · Pydantic	PostgreSQL / SQLite · Amazon DynamoDB	Amazon Bedrock (Nova Lite): interaction screening, clinical briefs	Cognito · S3 · KMS · SNS · SES · EventBridge · Step Functions · AppSync · CDK	argon2 · TOTP MFA · pytest · Vitest · GitHub

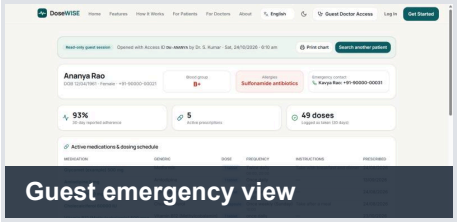
Implementation & Prototype



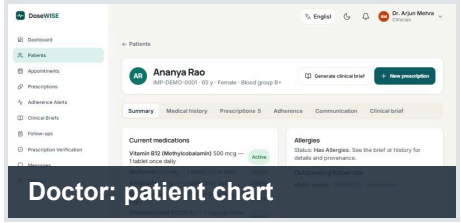
Patient dashboard: today's doses, Log dosage, Access ID, appointments and doctor instructions

Key Workflow

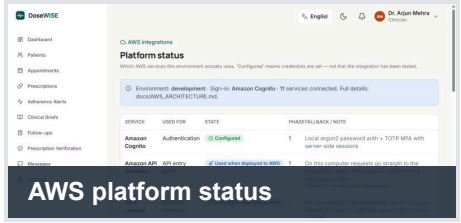
1. Doctor prescribes; safety checks run (duplicates, allergies, Bedrock AI)
2. Patient gets the reminder and logs the dose
3. Doctor and caregiver see taken / late / missed live
4. Emergency doctor opens a read-only chart with the Access ID (logged; patient notified)



Guest emergency view



Doctor: patient chart



AWS platform status

Current Implementation Status

Working prototype: 5 role portals, Cognito sign-up with email code, 11 AWS services connected, 5 languages (EN · HI · KN · TA · TE)

Feasibility & Impact

Feasibility

- Already built and running on managed AWS services in Mumbai; no special hardware
- Works in any browser on phone or laptop
- Dependencies: a licensed drug-interaction database (AI screening is marked unverified until then) and TRAI DLT registration for SMS

Validation

DEMONSTRATED

- 51 automated tests pass (39 backend, 12 frontend)
- All 114 frontend API calls verified against backend routes; 51 pages across 5 roles opened with 0 errors
- Live on AWS: Cognito sign-up with real email code; S3 encrypted storage; Bedrock flagged warfarin + aspirin

Potential Impact

PROJECTED

- Fewer missed doses and earlier follow-up by doctors and caregivers
- Safer prescribing when several doctors are involved
- Emergency doctors see medicines and allergies in seconds
- A 30-patient pilot would measure adherence against the WHO ~50% baseline

Key Metrics

Measured on the prototype on 24/10/2026; cost is an estimate

AWS services live

11 services

Measured · *Platform status*

Automated tests passing

51 tests

Measured · *pytest + Vitest*

Pilot cloud cost (1k patients)

$70–95 / month

Projected · *estimate, SMS off*

Roadmap & Scalability



Immediate Next Steps

- Deploy the CDK stack (API Gateway, Lambda, RDS)
- Register with TRAI DLT and switch on SMS reminders
- License a drug-interaction database
- Clinical review and DPDP legal review

Scalability

- Serverless compute and managed databases grow with demand
- Known limits, planned fixes: DB connections → RDS Proxy; reminder engine → SQS; live updates → AppSync
- Estimated $360–720 / month at 50k patients

Future Development

- Android / iOS app with offline reminders
- ABDM and FHIR (Amazon HealthLake) record exchange
- Refill reminders and nearby pharmacies via Location Service
- Adherence analytics for doctors; more Indian languages